FinTech 101

Payments

Intro

Since money and payments are interconnected you will see many information that you are already familiar with from the “History of Money” lecture, that is why we will just briefly look at the history of payments and will move on to the payment standards and the future of payments.

History: Pre-Banking

- **Barter System** (Prehistoric Times to Ancient Civilizations)
- **Commodity Money** (Ancient Times)
- **Metal Coins** (7th Century BC - Middle Ages)
- **Paper Money** (7th Century AD - Modern Era)

History: Banking

Banking Systems (Medieval Europe onwards): Banks emerged as institutions for storing money, providing loans, and facilitating payments.

Checks (17th Century): Checks became a popular method of payment, allowing individuals to transfer funds without the need for physical currency.

Credit Cards (20th Century): Credit cards revolutionized payment systems by allowing consumers to make purchases on credit. Initially introduced by individual stores and later by banks, credit cards became ubiquitous in modern economies.

Electronic and Digital Payments

Electronic Payments (Late 20th Century - Present): The rise of computers and the internet led to the development of electronic payment systems.

Digital and Mobile Payments (21st Century): With the proliferation of smartphones and internet connectivity, digital and mobile payment platforms have become increasingly popular.

Cryptocurrencies (2009 - Present): Bitcoin, introduced by an unknown person or group using the pseudonym Satoshi Nakamoto, marked the beginning of cryptocurrencies.

Payment Rails

Whenever we make a payment, an information exchange about your payment is happening between different financial institutions.

Payment rails refer to the underlying systems and networks that facilitate the transfer of funds between parties during a financial transaction. These systems serve as the infrastructure that enables the movement of money from one account to another.

The specific payment rails available vary by country and can include various electronic funds transfer mechanisms, card networks, and digital payment systems.

Payment Rails: US

- **Automated Clearing House (ACH):** ACH is a network for electronic payments and transfers in the United States, used for direct deposits, bill payments, and other transactions. It's run by NACHA and is usually cheaper than a wire transfer.
- **Fedwire:** Fedwire is a real-time gross settlement system operated by the Federal Reserve Banks, enabling immediate transfer of funds between financial institutions.
- **The Clearing House (TCH):** TCH operates systems for processing large-value and time-sensitive payments, including the CHIPS (Clearing House Interbank Payments System) for high-value transactions.

Payment Rails: UK

- **Faster Payments Service (FPS)**: is a near-real-time payment system in the UK, facilitating electronic transfers between bank accounts within seconds. Created in 2008, the Faster Payment System is available 24/7/365 to most customers in the UK with a bank account.
- **Bacs**: Bacs Payment Schemes Limited operates the Bacs system for processing bulk electronic payments, including direct debits and credits. BACS ensures that financial transactions flow from one bank account to another throughout the UK.
- **CHAPS (Clearing House Automated Payment System)**: CHAPS is a same-day sterling payment system for high-value transactions in the UK. It involves the settlement of foreign exchange and money market transactions by financial institutions, businesses and individuals.

Payment Rails: EU

- **SEPA (Single Euro Payments Area):** SEPA allows for the integration of electronic payments across EU member states, enabling cross-border transfers in euros under the same conditions as domestic payments. This local EU payment rail provides the same procedure for making both national and international payments.
- **TARGET2:** is the Trans-European Automated Real-time Gross Settlement Express Transfer System owned and operated by the Eurosystem. Central banks and commercial banks can submit payment orders in euro to TARGET2, where they are processed and settled in central bank money, i.e. money held in an account with a central bank.

Payment Rails: China

- **China National Advanced Payment System (CNAPS)**: CNAPS is China's interbank payment and settlement system, facilitating electronic funds transfers and settlements among banks. CNAPS is operated by the China National Clearing Center (CNCC) and local clearing systems are mainly operated by PBC branches.
- **UnionPay**: UnionPay (also known as China UnionPay, CUP or UPI) is the world’s biggest card network with more than 7 billion cards issued. It is widely used by Chinese shoppers internationally as well as domestically.

Payment Rails: India

- **Unified Payments Interface (UPI)**: UPI is a real-time payment system in India that enables instant fund transfers between bank accounts using mobile phones. With UPI, users can access multiple bank accounts in a single mobile application, allowing them to move money in real time.
- **National Electronic Funds Transfer (NEFT)** and **Real-Time Gross Settlement (RTGS)**: NEFT and RTGS are electronic payment systems in India, used for transferring funds between accounts held in different banks.
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Payment Rails: SWIFT

SWIFT - Society for Worldwide Interbank Financial Telecommunication.

SWIFT was formed in 1973 to help standardize international funds transfers and prevent security risks that traditional telegraphic transfers posed. Its inaugural year, it was initially supported by 239 banks in 15 countries. By the time the first message was sent in 1977, that had expanded to 518 banks in 22 countries.

SWIFT is the network of 11,000 member financial institutions that enables fast, secure international wire transfers. The network doesn’t transfer any physical funds, but rather provides a standard way to send payment orders. In 2020, around 35 million orders per day were sent via the SWIFT network.

Credit Card Networks

Credit card networks provide the communication system that issuing banks and businesses use to process credit card transactions.

There are two types of credit card networks – open and closed – and they treat card issuing differently.

Open credit card networks allow other financial institutions to issue their credit cards to customers:

- **Visa**
- **Mastercard**

In **closed credit card networks**, the credit card company exclusively issues the cards**:**

- **American Express**
- **Discover**

Credit card networks work by connecting the two parties involved in credit card purchases: the card issuer and the business.

Digital and Mobile Payments

With the proliferation of smartphones and internet connectivity, digital and mobile payment platforms have become increasingly popular. Services like PayPal, Venmo, Apple Pay, Google Pay, and cryptocurrencies offer convenient and secure ways to send and receive money electronically.

These types of mobile payments are still build on top of the existing banking or credit card payments infrastructure but add a layer of convenience on top.

Cryptocurrencies

Bitcoin, introduced by an unknown person or group using the pseudonym Satoshi Nakamoto, marked the beginning of cryptocurrencies.

Built on blockchain technology, cryptocurrencies aim to provide decentralized, secure, and anonymous transactions, challenging traditional financial systems.

Bitcoin and other cryptocurrencies that used for payments are built on top of the entirely new blockchain based infrastructure.

What comes next

FinTech

RegTech

SupTech

InsurTech

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